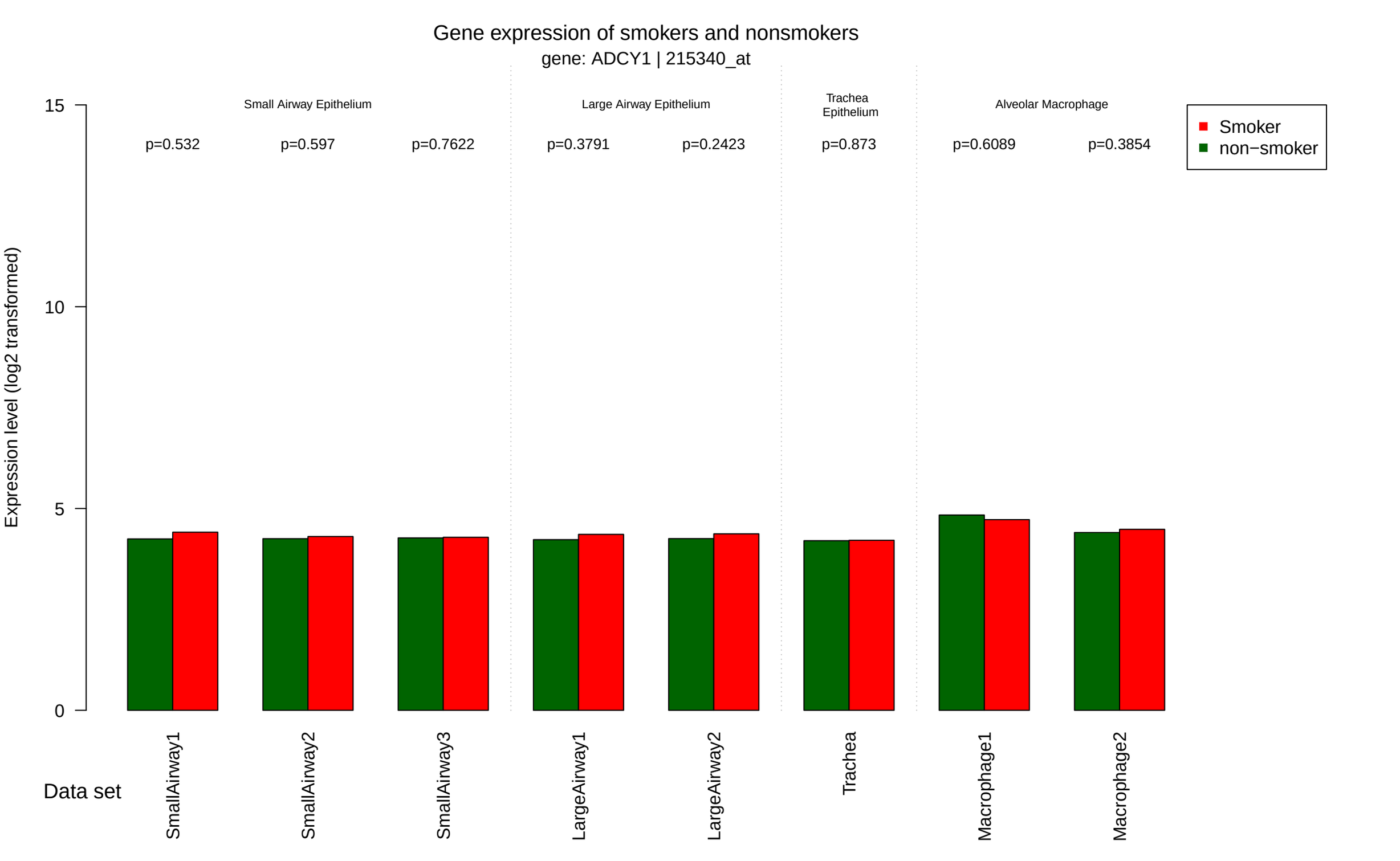
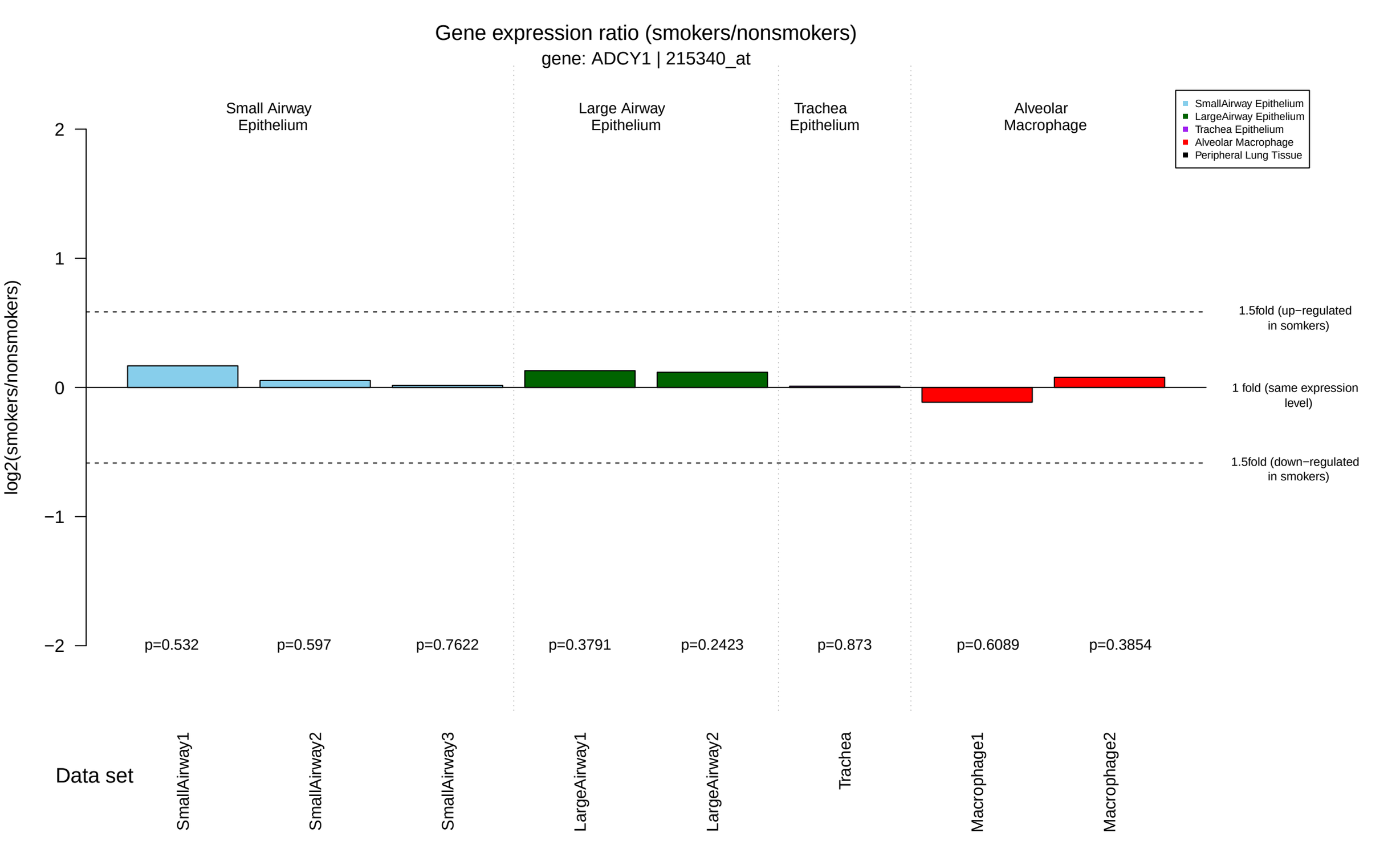
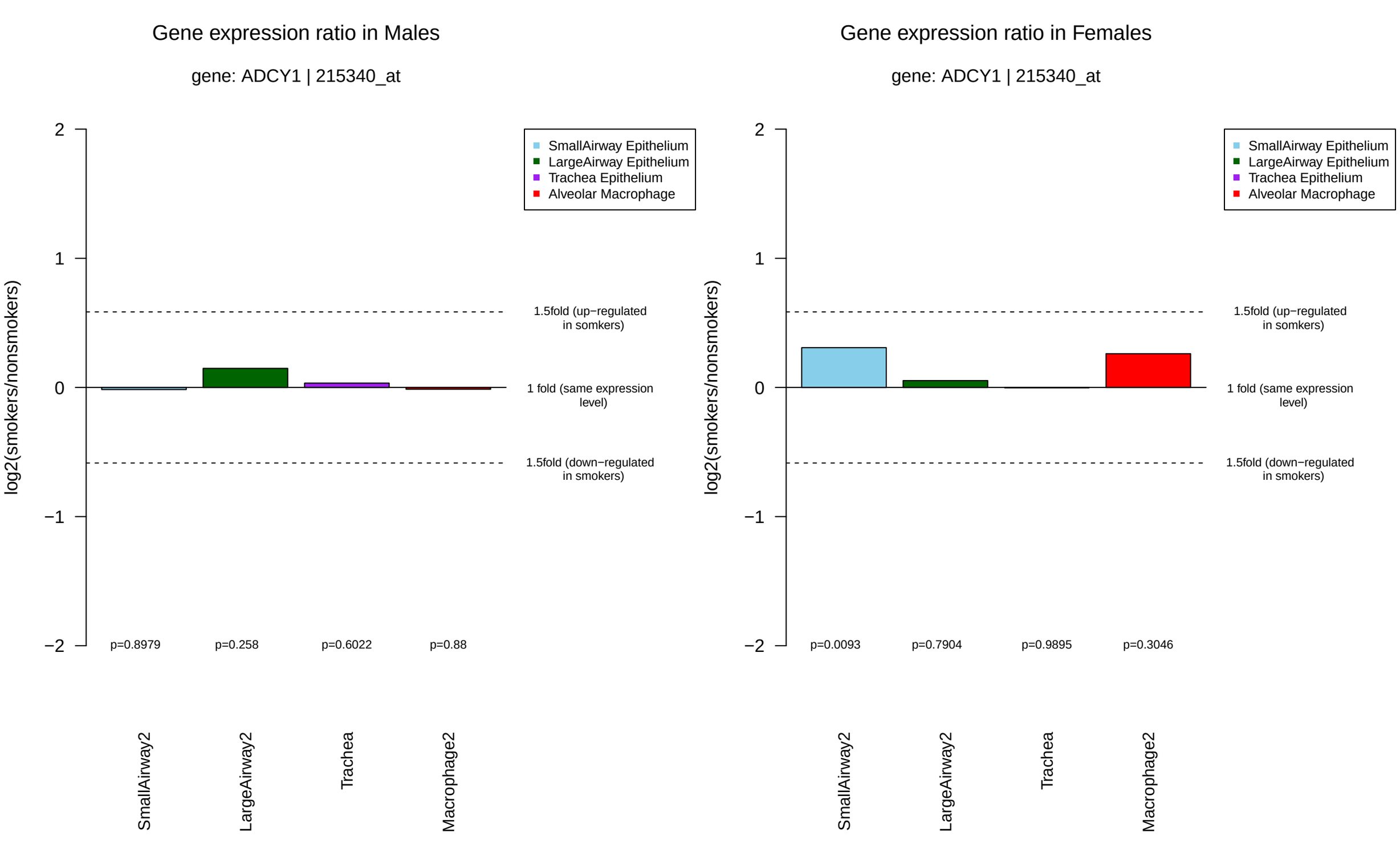
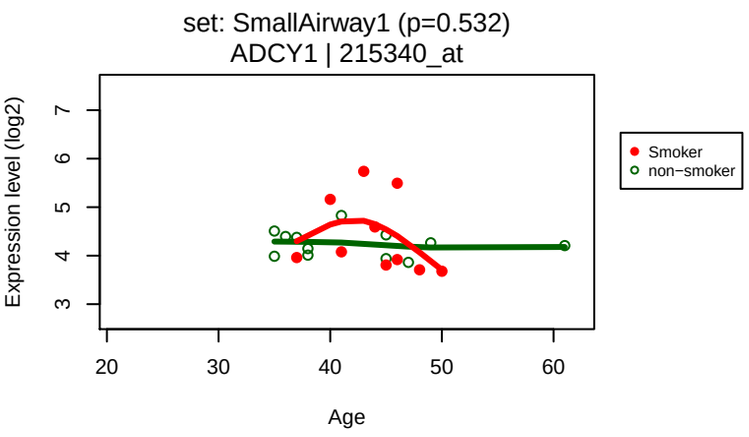
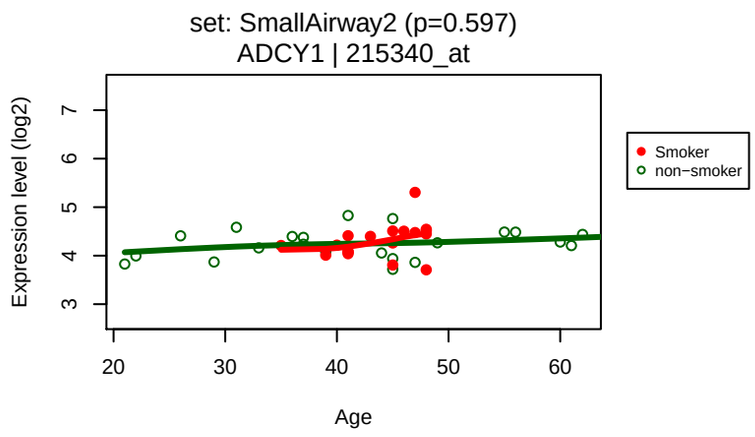
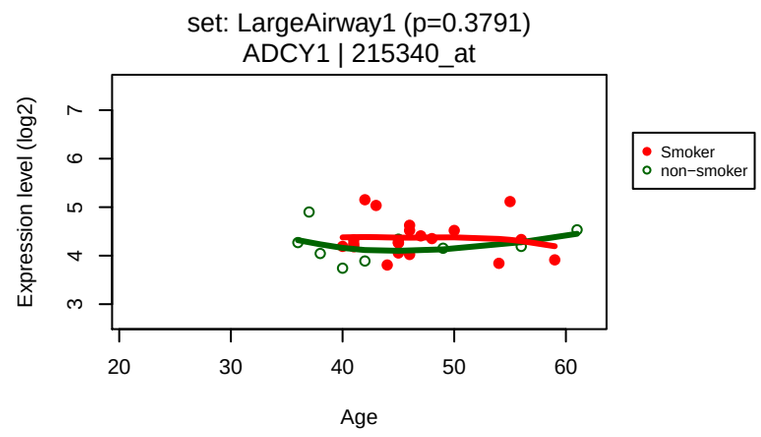

Gene expression of smokers and nonsmokers
gene: ADCY1 | 215340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.532
p=0.597
p=0.7622
p=0.3791
p=0.2423
p=0.873
p=0.6089
p=0.3854
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

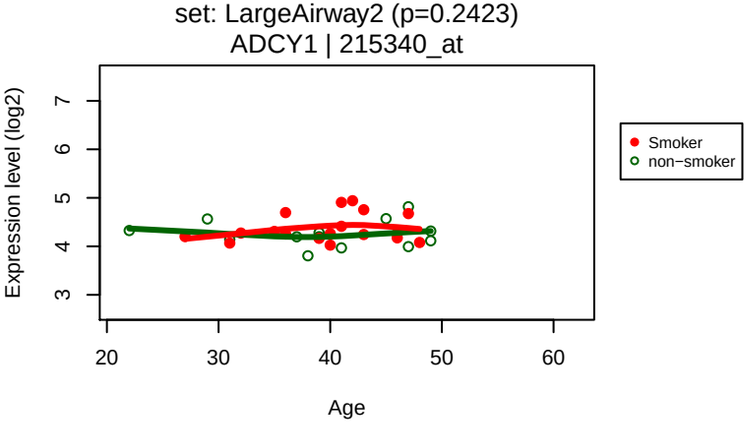

Gene expression ratio (smokers/nonsmokers)
gene: ADCY1 | 215340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.532
p=0.597
p=0.7622
p=0.3791
p=0.2423
p=0.873
p=0.6089
p=0.3854
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

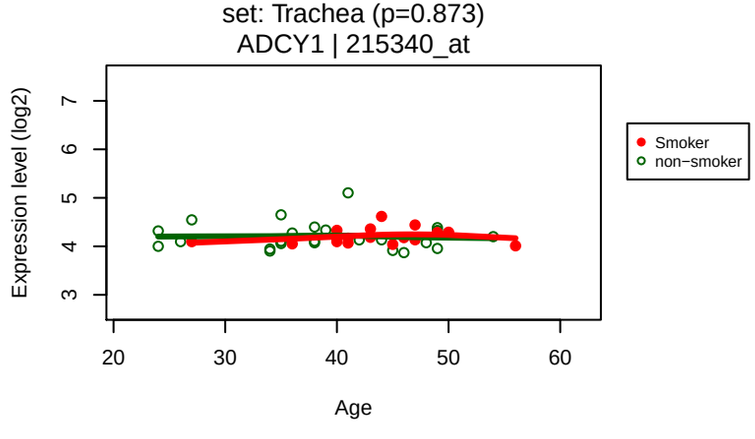

Gene expression ratio in Males
gene: ADCY1 | 215340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8979
p=0.258
p=0.6022
p=0.88
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY1 | 215340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0093
p=0.7904
p=0.9895
p=0.3046
SmallAirway2
LargeAirway2
Trachea
Macrophage2

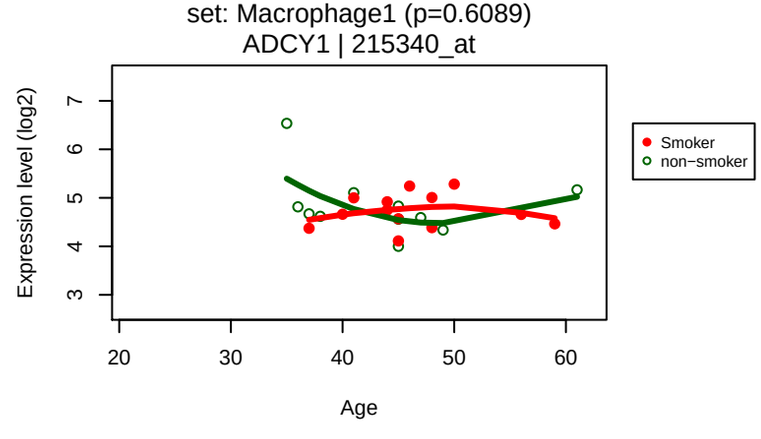

set: SmallAirway1 (p=0.532)
ADCY1 | 215340_at
Expression level (log2)
Age
Smoker
non−smoker

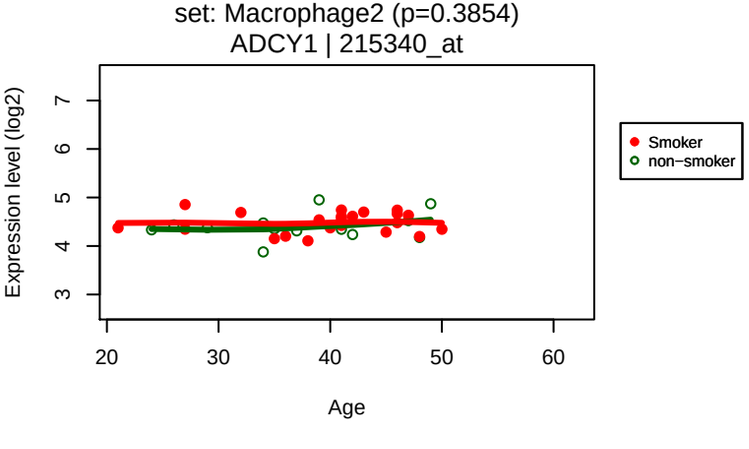

set: SmallAirway2 (p=0.597)
ADCY1 | 215340_at
Expression level (log2)
Age
Smoker
non−smoker

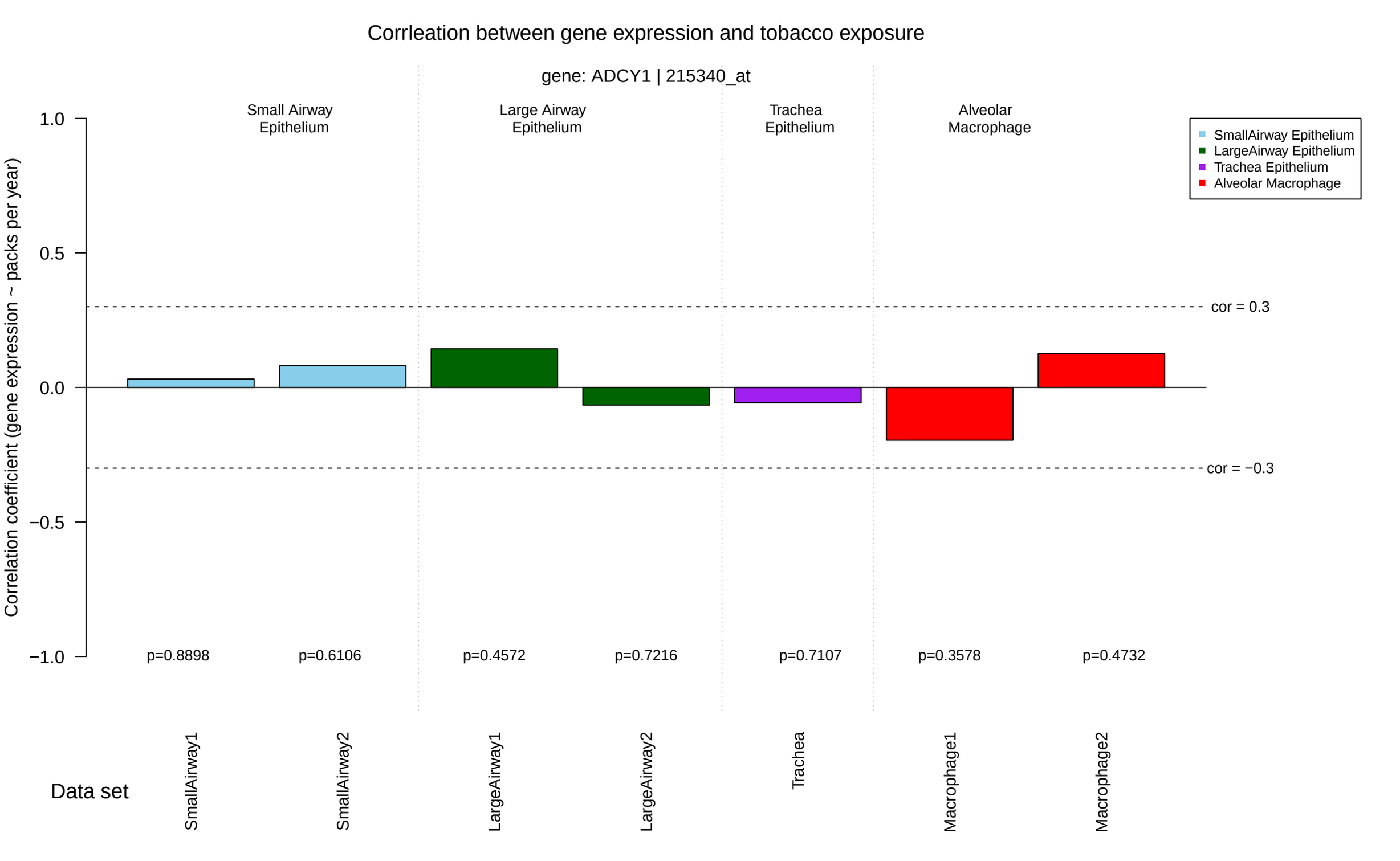

set: LargeAirway1 (p=0.3791)
ADCY1 | 215340_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2423)
ADCY1 | 215340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.873)
ADCY1 | 215340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6089)
ADCY1 | 215340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3854)
ADCY1 | 215340_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY1 | 215340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8898
p=0.6106
p=0.4572
p=0.7216
p=0.7107
p=0.3578
p=0.4732
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2